

Publication Bias in Measuring Anthropogenic Climate Change

Dominika Reckova, Zuzana Irsova

Institute of Economic Studies
Faculty of Social Sciences
Charles University in Prague

Outline

- 1 Introduction
- 2 Climate sensitivity
- 3 Data
- 4 Methodology
- 5 Results
- 6 Conclusion

- MRA of the relation between the concentration of carbon dioxide in the atmosphere and changes in global temperature
- Climate sensitivity
- 48 estimates from 16 studies
- Evidence for publication bias in 14 models
- No such analysis has previously been published

Climate Sensitivity

- “defined as the response in global-mean near-surface temperature to a doubling of atmospheric CO₂ concentrations from preindustrial levels” (Klocke *at al.* 2011)
- Also other factors influence the temperature change. (Edwards et al., 2007)
- The third assessment report of the IPCC predicts that CS likely ranges between 2 and 4.5 and is very unlikely to be less than 1.5.
- The fifth assessment report of the IPCC predicts that CS probably ranges from 1.5 to 4.5 with high confidence and is extremely unlikely to be lower than 1.
- Michaels (2008) uses vote-counting to analyse 116 issues of two journals that forecast climate change: *Science* and *Nature* and he found bias towards “worse” results.

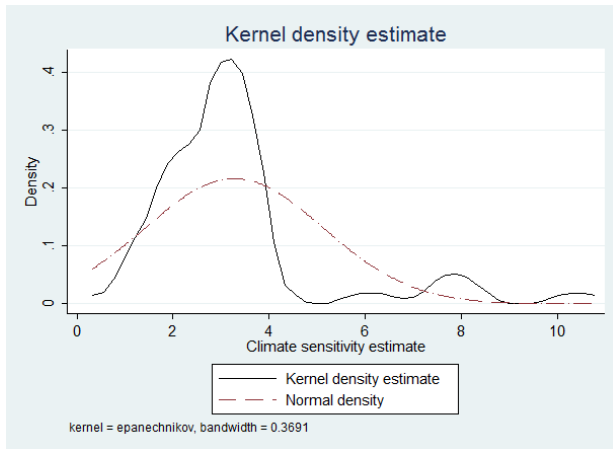


Figure 1: The kernel density of climate sensitivity estimates

- Searching for data till May 13, 2014 in Google Scholar and Web of Knowledge.
- 16 published papers provide 48 estimates of climate sensitivity (mostly mean or median).
- These range from 0.7 to 10.4, with an average of 3.27 °C.
- In the absence of bias the estimates are normally distributed around the hypothetical true effect.

Funnel plot of the estimated CS

- Calculating the Standard Errors:

$$SE_{low} = \frac{estimate - lower_bound}{z} \quad (1)$$

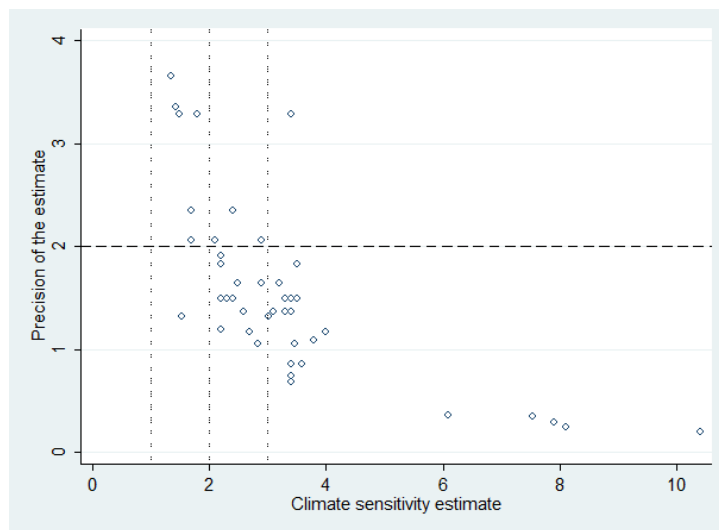


Figure 2: Funnel plot of the estimated CS

Note: This figure excludes the single most precise estimate to zoom in on the relationship.

Response variable: t-statistic	ME	Clus. OLS	Clus. FE
1/SE	1.689*** (0.188)	1.425*** (0.3)	2.15*** (0.085)
mean/SE	-1.105*** (0.186)	-0.832*** (0.274)	-1.573*** (0.077)
Constant (bias)	2.192*** (0.328)	2.577*** (0.178)	2.043*** (0.08)
Observations	48	48	48
Likelihood-ratio test (χ^2)	8.82***		
R^2		0.707	0.637

Table 1: Test of publication bias

- Doucouliagos & Stanley (2013) regard a FAT result of higher than 2 in absolute terms as “severe” selectivity.
- 14 models indicate strong publication bias at least at 5% significance level.
- The estimates of the true CS range between 1.4 and 2.3 in the extreme cases, and the average is 1.74.

Response variable: t-statistic	ME	Clus. OLS	Clus. FE
1/SE (true CS)	1.617*** (0.19)	1.276*** (0.316)	2.087*** (0.086)
mean/SE	-1.074*** (0.183)	-0.732** (0.286)	-1.55*** (0.079)
SE	-0.234* (0.132)	-0.316*** (0.086)	-0.226*** (0.017)
Constant (bias)	2.5*** (0.369)	3.054*** (0.232)	2.353*** (0.068)
Observations	48	48	48
Likelihood-ratio test (χ^2)	8***		
R^2		0.728	0.647

Table 2: Test of true climate sensitivity beyond publication bias

Conclusion

- CS estimates play a crucial role in evaluating the impacts of anthropogenic climate change
- One of the most important inputs into the computation of the social cost of carbon
- Evidence for publication selection bias: researchers tend to report preferentially large estimates of climate sensitivity
- Corrected for publication bias, the bulk of the literature is consistent with climate sensitivity lying between 1.4 and 2.3°C

References



Doucouliagos, C. & T. Stanley (2013)

Are all economic facts greatly exaggerated? theory competition and selectivity.
Journal of Economic Surveys 27(2), pp.316 – 339.



Klocke, D., R.Pincus, & J. Quaas (2011)

On constraining estimates of climate sensitivity with present-day observations through model weighting.
Journal of Climate 24(23), pp. 6092.

Thank you for your attention!